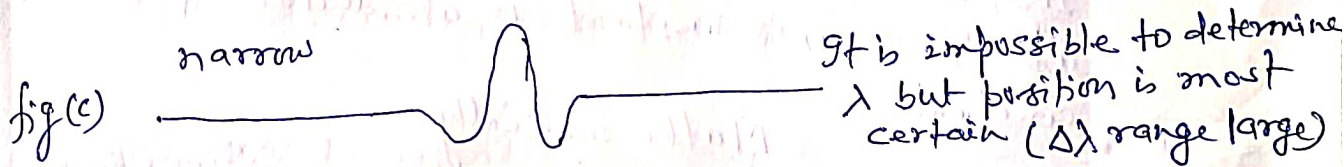
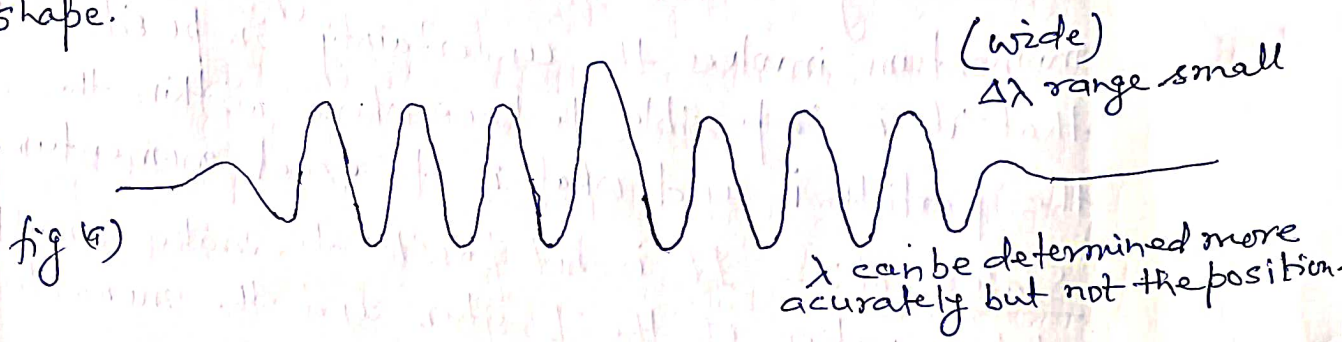


UNCERTAINTY PRINCIPLE

In 1927, Heisenberg proposed a very interesting principle which is a direct consequence of the dual nature of matter, known as uncertainty principle. The fact that a moving body must be regarded as a de-Broglie wave group rather than as a localized entity suggests that there is a fundamental limit to the accuracy with which its particle properties can be measured. The wave packet can be of any shape.



If the constituent waves are small in number ($\Delta\lambda$ range is small), then the wave packet is wide because over a large length constructive interference takes place (fig a). The wave packet is narrow, when the constituent waves are more in number. The greater the range of propagation constant, the narrower the wave packet. In a wave packet, the constituent waves have amplitude which vary with the likelihood of finding/detecting the particle. The greater the peak of amplitude, the greater the probability of finding the particle there.

If the wave packet is narrow as in fig(c), the position of the particle is readily found, but the wavelength is impossible to establish & hence the momentum ($\lambda = \frac{h}{p}$). On the otherhand, if the wave group is wide, wavelength can be estimated more accurately but the position of the particle becomes more uncertain. It is difficult to locate the particle exactly in a wide wave group.

In this way the certainty in position involves uncertainty in momentum or velocity and certainty of momentum involves the uncertainty in position. This shows that it is impossible to know where within the wave packet the particle is and, what is its exact momentum.

Statement: - It is not possible to make simultaneous determination of the position and the momentum of a particle with unlimited precision.

It further states that the product of the uncertainty Δx in the position of a body at some instant and the uncertainty Δp in the momentum at the same instant is at least equal to Planck's constant.

Mathematically

$$\boxed{\Delta x \Delta p \geq \hbar}$$

We cannot measure simultaneously, both position and momentum with perfect accuracy.

The other forms are

$$\Delta E \Delta t \geq \hbar$$

$$\text{and } \Delta J \Delta \theta \geq \hbar$$

ΔE & Δt are uncertainties in determining energy and time while ΔJ and $\Delta \theta$ are uncertainties in determining the angular momentum and angular displacement.

Derivation on the basis of Wave Nature of Particle

Let us consider the formation of wave group due to two ~~waves~~ de-Broglie wave given by

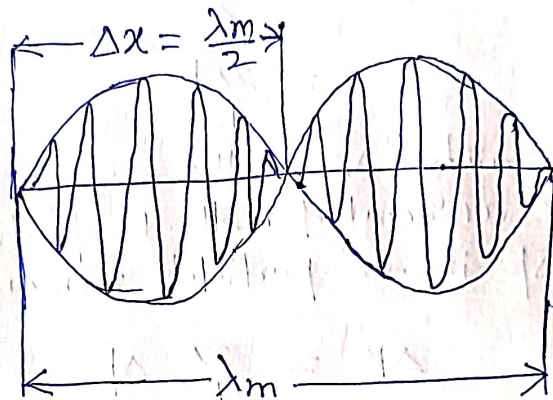
$$\psi_1 = A \cos(\omega t - kx)$$

$$\psi_2 = A \cos[(\omega + \Delta\omega)t - (k + \Delta k)x]$$

Hence

$$\psi = \psi_1 + \psi_2$$

$$= 2A \cos(\omega t - kx) \cos\left(\frac{\Delta\omega}{2}t - \frac{\Delta k}{2}x\right) \quad \text{--- (1)}$$



Width of each group = $\frac{\lambda_m}{2}$, where λ_m = wavelength of the modulated wave. Hence uncertainty in the position of particle

$$\Delta x \approx \frac{\lambda_m}{2} \quad \text{--- (2)}$$

If k_m be the propagation constant of the modulated wave, then

$$\lambda_m = \frac{2\pi}{k_m} \quad \text{--- (3)}$$

From eqⁿ (1), we have

$$k_m = \frac{\Delta k}{2} \quad \text{--- (4)}$$

Hence equation (3) reduces to

$$\lambda_m = \frac{2\pi}{\Delta k} \times 2 \quad \text{--- (5)}$$

Putting this value of λ_m in equation (2), we have

$$\Delta x = \frac{2\pi}{\Delta k} \times \frac{2}{2}$$

$$= \frac{2\pi}{\Delta k} \quad \text{--- (6)}$$

The constituent waves have propagation constant K and $K + \Delta K$. ΔK can be taken as the inherent uncertainty in the best measurement of K . Hence the uncertainty ΔK will be given by

$$\Delta K = \frac{2\pi}{\Delta x} \quad \text{--- (7)}$$

The de-Broglie wavelength of a moving particle of momentum p is given by

$$\lambda = \frac{h}{p}$$

The propagation constant corresponding to this wavelength is

$$K = \frac{2\pi}{\lambda} \\ = \frac{2\pi p}{h}$$

Hence uncertainty ΔK in the propagation constant of the de-Broglie waves associated with the particle results in an uncertainty Δp in the particle's momentum given by

$$\Delta K = \frac{2\pi}{h} \Delta p \quad \text{--- (8)}$$

From eqⁿ (7) and (8), we have

$$\frac{2\pi}{\Delta x} = \frac{2\pi}{h} \Delta p$$

$$\text{or } \boxed{\Delta x \Delta p = h}$$

In the analysis only two superposed wave trains have been considered. But two superposed wave trains yield a series of groups as shown. To achieve a single wavegroup an infinite number of wave trains whose frequencies and hence propagation constants K differ only infinitesimally from one another is required. The narrower the group, the greater the range of propagation constant that is needed. Fourier analysis shows that

$$\Delta x \geq \frac{1}{\Delta K}$$

Hence eqⁿ (8) reduces to

$$\frac{2\pi}{h} \Delta p = \frac{1}{\Delta x}$$

$$\text{or } \Delta x \Delta p = \frac{h}{2\pi}$$

$$\text{or } \boxed{\Delta x \Delta p \geq \hbar}$$

Thus it is clear that we cannot measure simultaneously both position and momentum with perfect accuracy. If we arrange matters so that Δx is small corresponding to the narrow wave group, Δp will be large. If we reduce Δp in some way corresponding to the wide wave group, Δx will be large. These uncertainties are not in our apparatus but in nature.

Alternate Form of Heisenberg's Principle

Suppose one wish to measure the energy E emitted sometime during the time interval Δt in an atomic process. If the energy is in the form of e.m. waves, the limited time available restricts the accuracy with which the frequency ν of the waves can be determined. Let us assume that the irreducible error in the number of waves counted in a wavegroup is one wave. Since

$$\text{frequency } \nu = \frac{\text{no. of waves counted}}{\text{time interval}}$$

Therefore, uncertainty in frequency

$$\Delta \nu = \frac{1}{\Delta t}$$

Now

$$\text{Energy } E = h\nu$$

$$\Delta E = h \Delta \nu$$

$$= h \frac{1}{\Delta t}$$

$$\text{or } \Delta E \Delta t = h$$

A more realistic calculation shows that

$$\boxed{\Delta E \Delta t \geq h}$$

Another Derivation

$$\text{velocity of the moving particle } v = \frac{\Delta x}{\Delta t}$$

$$\text{or } \Delta x = v \Delta t$$

$$\text{Kinetic energy } E = \frac{p^2}{2m}$$

$$\Delta E = \frac{2p \Delta p}{2m}$$

$$= v \Delta p$$

$$= \frac{\Delta x \Delta p}{\Delta t}$$

$$\text{or } \Delta E \cdot \Delta t = \Delta x \Delta p$$

$$\text{Since } \Delta x \Delta p \geq \hbar$$

$$\text{So, } \boxed{\Delta E \Delta t \geq \hbar}$$

Statement: - It is not possible to make a simultaneous determination of the energy and the time co-ordinate of a particle with unlimited precision.

Example: Let us consider the radiation of light from an excited atom. An atom takes 10^{-8} sec in coming from excited state to ground state. The uncertainty in the measurement $\Delta t = 10^{-8}$ s

uncertainty in energy

$$\Delta E = \frac{\hbar}{\Delta t}$$

$$= \frac{1.054 \times 10^{-34} \text{ J}}{10^{-8}}$$

$$= 1.1 \times 10^{-26} \text{ J}$$

The frequency of the light is uncertain by

$$\Delta E = h \Delta \nu$$

$$\Delta \nu = \frac{\Delta E}{h}$$

$$= \frac{1.1 \times 10^{-26}}{6.6 \times 10^{-34}} \text{ cycles/s}$$

$$= 1.6 \times 10^7 \text{ cycles/s}$$

This is the irreducible limit to the accuracy with which we can measure the frequency of radiation emitted by an atom.